

DS Unit 5 Previous Year Questions

Data Structures (DS) · Unit 5 · Sorting, Searching & Hashing · 2023

Questions collected from previous year university examinations (2019–2023).

Part A — Short answer (2 marks)

1. Write short notes on Binary Search. [2021]
2. Discuss the role of Quick Sort in Sorting, Searching & Hashing. [2020]
3. Explain Merge Sort with a neat diagram and a suitable example. [2022]
4. Explain Heap Sort with a neat diagram and a suitable example. [2022]

Part B — Long answer (10 marks)

1. Discuss the role of Hash Functions in Sorting, Searching & Hashing. [2021]
2. What is meant by Collision Resolution? Discuss its advantages and limitations. [2021]
3. Define Binary Search. Derive or justify the key properties involved. [2022]
4. Explain Quick Sort with a neat diagram and a suitable example. [2022]
5. What is meant by Merge Sort? Discuss its advantages and limitations. [2023]
6. Illustrate Heap Sort with a worked example. [2022]
7. Compare Hash Functions with Collision Resolution. When would you prefer one over the other? [2022]
8. Define Collision Resolution. Derive or justify the key properties involved. [2022]
9. Explain Binary Search with a neat diagram and a suitable example. [2023]
10. Illustrate Quick Sort with a worked example. [2019]